Date **SIFT FEATURES**

Ex.Nos. 10

Aim

To explore the MATLAB programs and to extract SIFT features

Algorithm

1. Login into online MATLAB
2 Read the input image
3. If the input image is RGB convert to gray scale image
4. Detect SIFT features
5. Display detected SIFT strongest keypoints
6. Extract SIFT Descriptorst
7. Display the size and number of SIFT features

Program:

1.Original image

Code:

```
clc;
clear;
close all;
I = imread('cameraman.tif');
if size(I,3) == 3
Igray = rgb2gray(I);
else
Igray = I;
end
figure;
imshow(Igray);
title('Original Image');
```

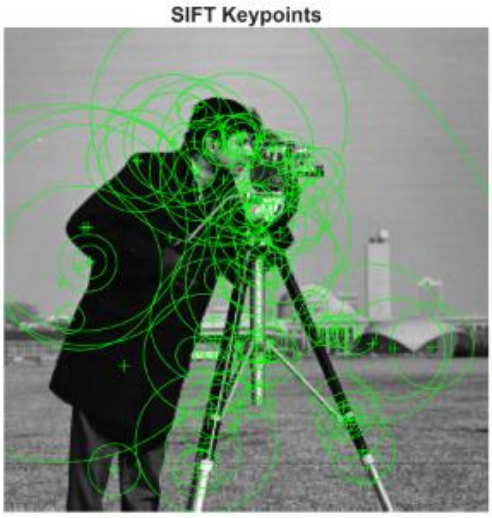
Output:

2.SIFT keypoints

code:

```
points = detectSIFTFeatures(Igray);
figure;
imshow(Igray);
hold on;
plot(points.selectStrongest(100));
title('SIFT Keypoints');
```

Output:



3.Size of SIFT

Code:

```
[features, validPoints] = extractFeatures(Igray, points);
disp('Number of SIFT keypoints:');
disp(validPoints.Count);
disp('Size of SIFT feature descriptor:');
disp(size(features));
```

Output:

```
Number of SIFT keypoints: 311
Size of SIFT feature descriptor: 311 128
```